



QuantumVault
PQC Enterprise Security by Dytallix

Quantum Risk Profile

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Understanding Quantum Threats

HNDL (Harvest Now, Decrypt Later)

Adversaries are collecting encrypted data today, storing it until quantum computers can decrypt it. Data that needs to remain confidential for 10+ years is already at risk.

CRQC (Cryptographically Relevant Quantum Computer)

A quantum computer powerful enough to break current encryption. Expected within 10-15 years, this will render RSA, ECC, and similar algorithms obsolete.

Shor's Algorithm

Breaks RSA and ECC by efficiently factoring large numbers. Eliminates public-key cryptography security.

Grover's Algorithm

Speeds up brute-force attacks on symmetric encryption. Reduces AES-256 to AES-128 equivalent security.

Your Organization Profile

Industry: Finance & Banking

Region: United States

Organization Size: Enterprise (500 - 5000 employees)

Regulatory Regime: SEC / NYDFS (US Finance)

Data Types: PII (Personally Identifiable Information) • Financial Records

Current Cryptography: RSA-2048 (Legacy) • AES-256 (Symmetric)

Your Quantum Risk Assessment

HNDL Risk



75 (High)

CRQC Risk



65 (Medium)

Migration Urgency



60 (Medium)

Post-Quantum Cryptography Transition Strategy

To protect your organization against quantum threats, we recommend the following strategic roadmap:

1. Cryptographic Inventory Audit

Catalog all cryptographic assets including certificates, keys, algorithms, and protocols in use across your infrastructure.

2. Identify Quantum-Vulnerable Systems

Prioritize systems based on data sensitivity, longevity requirements, and exposure to HNDL attacks.

3. Evaluate NIST-Approved PQC Algorithms

Assess CRYSTALS-Kyber (key encapsulation), CRYSTALS-Dilithium, FALCON, and SPHINCS+ (digital signatures) for your use cases.

4. Implement Hybrid Encryption

Deploy hybrid schemes combining classical and post-quantum algorithms during the transition period for backward compatibility.

5. Migrate Critical Systems First

Prioritize long-lived secrets, key exchange mechanisms, and systems handling sensitive data with extended confidentiality requirements.

6. Continuous Monitoring

Stay informed about quantum computing advancements and update migration timelines accordingly.

Ready to Quantum-Proof Your Organization?

Our team of post-quantum cryptography experts can help you assess your risk, develop a migration roadmap, and implement quantum-safe solutions tailored to your needs.

Contact us: hello@dytallix.com | dytallix.com/contact

